# Range-Visual-Inertial Odometry Prototype for Terrain-Relative State Estimation Over Non-Planar Mars Terrain

Sean Vellequette[#1]

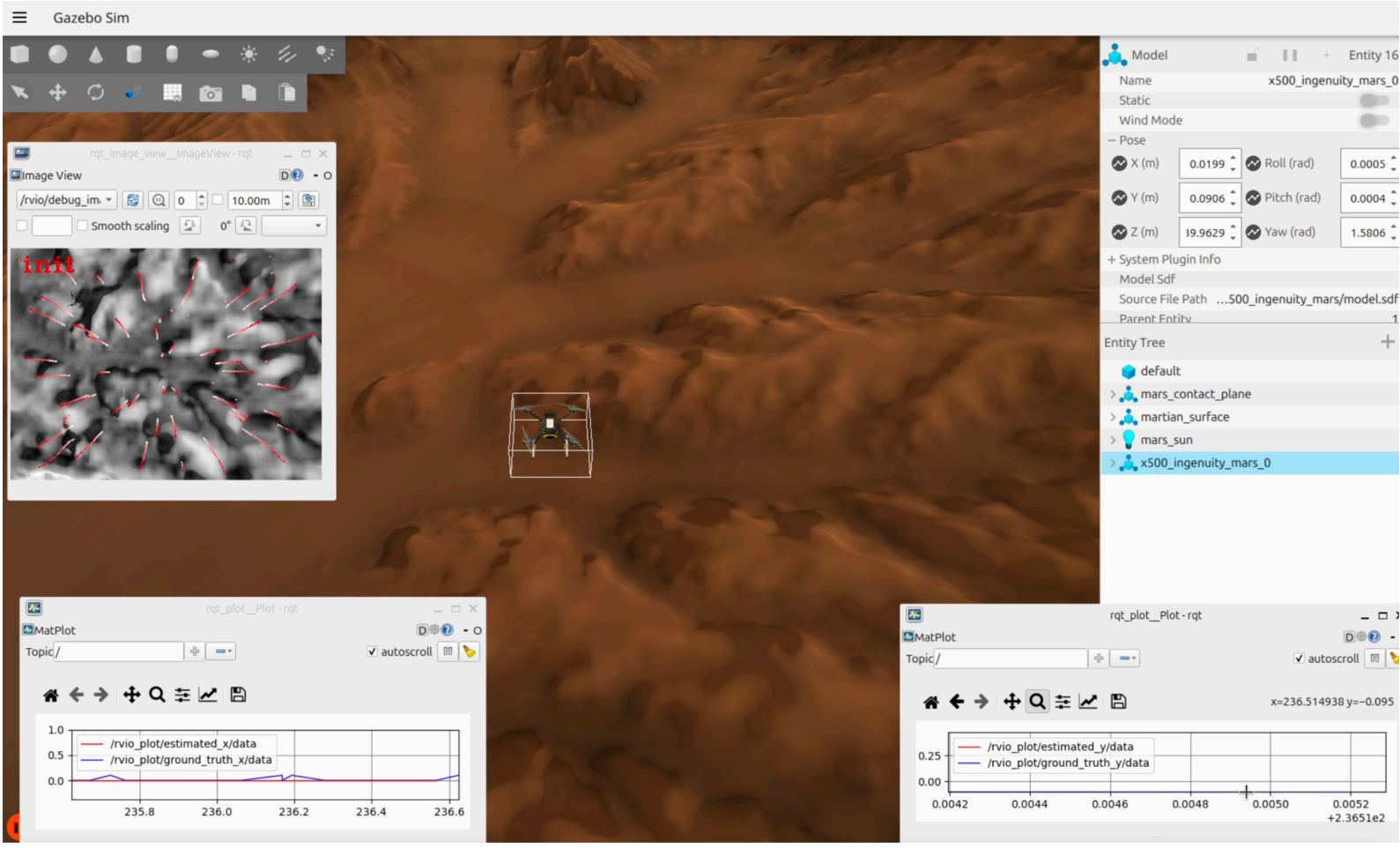
ASUID: 1222032539

[#]*Ira A. Fulton Schools of Engineering*

*Arizona State University*

*Mesa, AZ, USA*

[1]svellequ@asu.edu



***Abstract***—This project presents a proof-of-concept Range-Visual-Inertial Odometry (RVIO) prototype for a Mars-terrain drone mission in PX4/Gazebo. The project began from Assignment 3, the terrain_mapping_drone_control ROS 2 package, which originally supported an object-centric search, mapping, and landing workflow. For the final project, that package was reused as a scaffold and converted into a dedicated Mars RVIO evaluation pipeline. The new system focuses on terrain-relative state estimation rather than object landing. It uses a downward monocular camera, IMU data, and a one-dimensional downward lidar/rangefinder to estimate lateral motion over non-planar terrain, where monocular scale observability is weak and simple optical-flow assumptions are fragile. The sensor type used in the workflow are identical to the sensors on the original ingenuity drone. The final repository now runs PX4 SITL with the x500_ingenuity_mars model, Gazebo with mars_px4.sdf terrain model[1], an OpenVINS-based backend with range-feature support, a mission-aware delayed estimator launcher, a PX4 IMU bridge, and a terrain-relative evaluator with live plotting topics. The report also explicitly states the assumptions and caveats of the prototype, especially the fragility of one-dimensional range-feature association, camera/lidar alignment, timing sensitivity, and estimator initialization.

## 1. Introduction and Project Goal

This project investigates whether a downward monocular camera, IMU, and one-dimensional downward lidar can support terrain-relative state estimation for a Mars terrain drone in simulation. The prototype was motivated by the observation that visual navigation can become unreliable over structurally invariant terrain, particularly during descent or near-constant-velocity motion where monocular visual-inertial odometry can suffer from scale drift.

The central engineering question was intentionally narrow: can a custom built range-constrained visual-inertial estimator use sparse lidar information to improve local lateral velocity and lateral position estimation over non-planar terrain? The emphasis of this prototype is therefore on estimation, not on final landing decision logic. The current system does not claim a fully validated safe-landing capability. Instead, it establishes and evaluates the perception and estimation layer that would be required before a safe landing spot could later be selected.

This narrower scope was important for two reasons. First, the existing class assignment 3 infrastructure already flew a complete mission in PX4/Gazebo. Second, the most meaningful next step was not to redesign the entire mission stack, but to replace the simple planar or optical-flow reasoning with a stronger terrain-relative RVIO testbed and evaluation pipeline.

## 2. Prototype Objectives

The final project is best understood as a conversion and specialization of Assignment 3 rather than a completely separate codebase. The assignment package provided the PX4/Gazebo bridge scaffold. The final project added new estimation specific functionality, removed or de-emphasized assignment only logic, and recentered the mission around terrain relative localization over Mars terrain.

## 3. System Overview

The final system runs as a PX4 SITL and Gazebo Mars-terrain mission in ROS 2. The main mission launch starts PX4 SITL with a quadcopter airframe model, Gazebo with a Mars terrain world, the transport bridge between PX4 and ROS 2, the terrain-relative mission node, an OpenVINS-based RVIO backend, an evaluator node, and live plotting tools.

The vehicle carries a downward monocular camera, a downward 1D lidar/rangefinder, PX4 IMU data, and access to Gazebo ground-truth odometry for evaluation. This is reflected in the original Ingenuity drone [2]

The estimator outputs are exposed through ROS 2 topics including odometry, pose, path, and a debug image stream. The evaluator publishes live plotting topics for estimated and ground-truth local x, y, vx, and vy, along with aggregated metrics such as horizontal RMSE, velocity RMSE, and drift percentage. This instrumentation is a major deliverable of the project because it turns the simulation into a diagnostic testbed rather than only a visual demo

## 4. Mission Design and Air-Start Rational

An important practical lesson was that the estimator and evaluator should not begin collecting metrics at raw simulation spawn. Early tests showed that startup artifacts, terrain handoff issues, and transient range behavior polluted both the estimator and the metrics. At the same time, the mission design needed to remain faithful to the mission design described in the “Structure-Invariant Range-Visual-Inertial Odometry paper” [3]. Specifically the Powered Airborne Decent. The paper is not based on a standard ground takeoff mission. Instead, it describes a Mid-Air Helicopter Delivery (MAHD) concept in which the helicopter is already airborne after atmospheric entry and descent, and range-visual-interial navigation begins while the vehicle is above the terrain rather than parked on the ground.

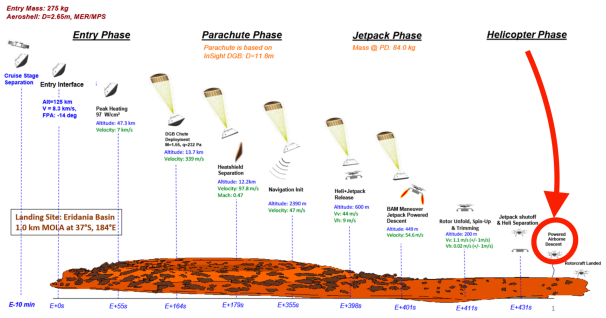



*Fig. 1: MAHD Helicopter EDL Phase*

The paper’s Figure 1 illustrates this mid-air deployment sequence, and the text explains that the navigation system must estimate terrain-relative velocity, orientation, and altitude during the airborne descent phase

For that reason, this project treated starting in the air above the terrain surface as an important modeling decision rather than just a software convenience. Spawning the drone already above the terrain allows the simulation to better approximate the paper’s mission concept, where the vehicle is not performing a conventional launch from the ground but instead begins its estimation problem already in an airborne descent context.

## 5. Ground Truth and Evaluation Frame

A key evaluation design choice was to compare all results in a local survey-start frame rather than directly in Gazebo’s raw world frame. This matters because the RVIO estimate is inherently local: it initializes in its own internal reference frame and then estimates relative motion from that origin. Gazebo’s world frame contains offsets that are not meaningful in this comparison.

To solve this, the evaluator locks the Gazebo pose when the mission enters SURVEY and treats that pose as the ground-truth origin. Ground-truth x and y are then published relative to this survey-start frame, and the RVIO estimate is compared against the same local convention. This allows the plots to answer the intended scientific question: from the beginning of the controlled survey segment, did the RVIO pipeline estimate the same local displacement and velocity as the simulator ground truth?

## 6. First Implementation: Python Proof of concept

The first RVIO implementation was a Python proof-of-concept node. Its purpose was fast development and rapid visibility into the sensor streams. It fused IMU propagation, optical flow from the downward camera, lidar-based scale cues, attitude compensation, and ground-truth diagnostics. This version was useful because it quickly exposed the live sensor interfaces and produced odometry and debug topics without requiring a full backend integration.

However, the proof of concept was also the point where the main geometric limitation became obvious. The simplified method behaved too much like an optical-flow shortcut. In particular, using a single lidar range value to scale visual motion is fragile when the tracked image feature is not actually on the lidar ray, when the terrain is non-planar, when the vehicle is pitched or rolled, or when the lidar range is treated too much like a vertical altitude measurement.

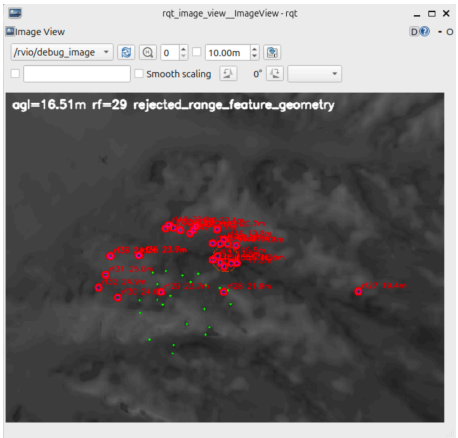



*Fig. 2: Range-Image Features (red) and Optical Flow (green)*

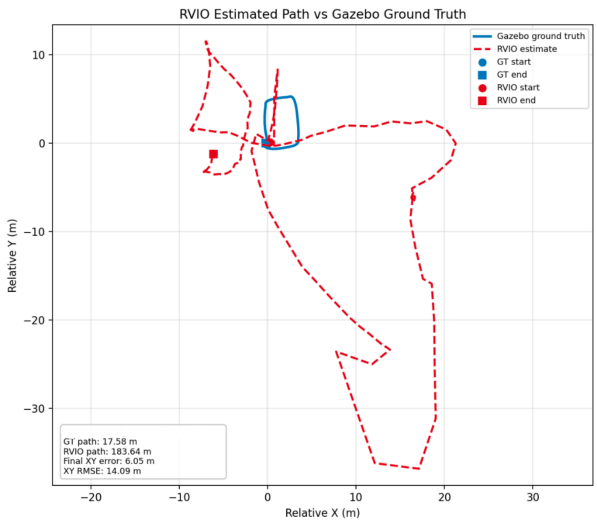



*Fig. 3: Initial X/Y RVIO OpenCV2 Estimation*

This was a useful result rather than a failure. It motivated the transition to a more realistic and established visual-inertial estimator structure[4].

## 7. OpenVINS-Based RVIO Integration

To get closer to a true visual-inertial odometry pipeline, the project integrated OpenVINS as the RVIO backend. This was a major architectural improvement because OpenVINS provides IMU propagation of a full 3D state, KLT feature tracking, camera-clone state handling, monocular update machinery, inverse-depth feature representations, and covariance propagation with Kalman updates. Not to mention that the paper Structure-Invariant Range-Visual-Inertial Odometry explicitly utilizes a hybrid Multi-State Constraint Kalman Filter for visual feature tracking.

The OpenVINS backend is launched in a mission-aware manner using a delayed launcher node. Instead of starting at simulation boot, OpenVINS begins when the mission reaches the SURVEY state. This isolates the evaluation window from early transients and makes it easier to interpret the resulting plots.

An additional IMU bridge node was required because PX4 publishes a message format different from what OpenVINS expects. This bridge converts PX4 sensor data into sensor_msgs/Imu and also mitigates time-base issues by re-anchoring PX4 IMU timing against the latest camera time. This step was necessary because the estimator is highly sensitive to image/IMU timing consistency.

## 8. OpenVINS-Based RVIO Results

At this stage, the preliminary results are mainly qualitative and systems-level, not a final quantitative benchmark study. The main success is that the simulation now closes the loop from mission execution to estimator output to local-ground-truth evaluation in real time. The prototype can fly a controlled Mars-terrain mission, start estimation during a deliberate survey phase, publish RVIO state estimates live, and compare local lateral motion against simulator ground truth.

The development process also produced an important preliminary finding. Moving to OpenVINS with sparse range-feature seeding is therefore a technically justified improvement, even though the final system remains a prototype and not a fully validated RVIO implementation.

In the current representation, the most useful plots are estimated local x versus ground-truth local x, estimated local y versus ground-truth local y, and the corresponding lateral velocity plots. These are the most direct diagnostics for the landing-risk question that motivated the work. If lateral motion cannot be mitigated, then later landing decisions are at risk even if the vehicle remains airborne and apparently stable.

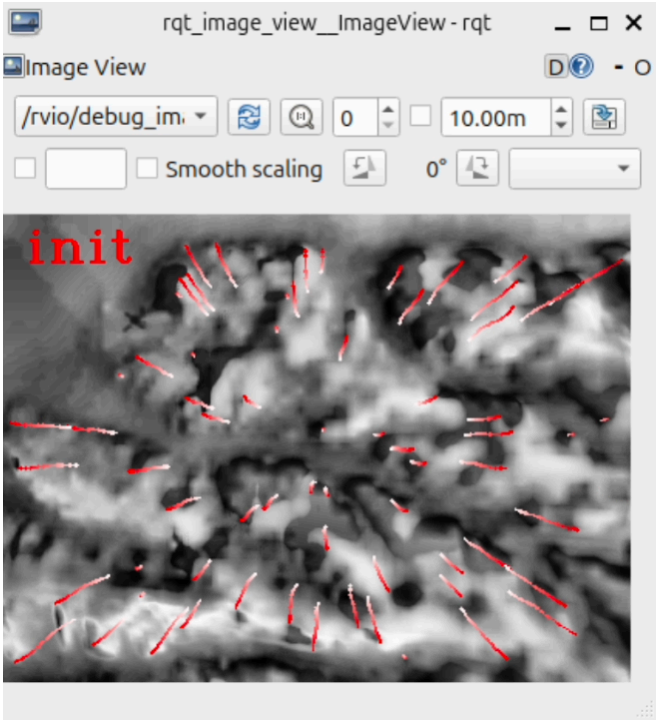



*Fig. 4: OpenVINS Optical Flow with Range-Features*

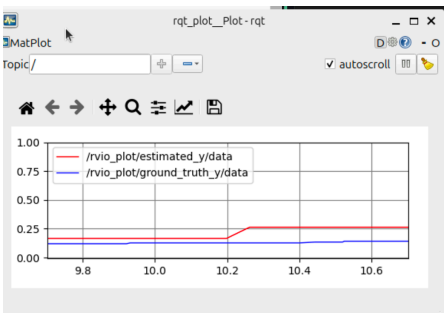



*Fig. 5: OpenVINS State Error Y During Movement*

## 9. Conclusions

This hybrid RVIO estimator produced promising outputs for terrain-relative state estimation over non-planar Mars-like terrain. The main result is not yet a complete safe-landing system, but a working estimation-and-evaluation pipeline that can compare local lateral position and velocity against simulator truth during controlled flight. The outputs suggest that combining visual-inertial estimation with sparse lidar-seeded range features is more promising than relying on a simpler optical-flow-style shortcut or treating a 1D downward lidar as a generic altimeter. This is consistent with the paper's main conclusion that range-feature initialization improves scale observability and terrain-relative velocity estimation without relying on planar terrain assumptions.

The most important caveat is that the current estimator remains a prototype. Its performance still depends on approximate camera-lidar alignment, accurate timing, sufficient visual feature quality, and correct association/transform between the lidar ray and the selected image feature. Because the lidar is only one-dimensional, its contribution is sparse and **can be fragile** over difficult terrain or weak-texture scenes. Even with those caveats, the current outputs support the conclusion that a hybrid RVIO approach is a promising direction for terrain-relative estimation in MAHD-like Mars flight scenarios.

## 10. Future Work

A good next step would be to measure the health of range-features before trusting them as metric constraints in the estimator. In the current hybrid RVIO design, a 1D lidar measurement is useful only when it is associated with the correct tracked image feature near the lidar ray. The paper's method depends on that range-feature being initialized with accurate depth and then improving the rest of the estimate through covariance propagation in the EKF.

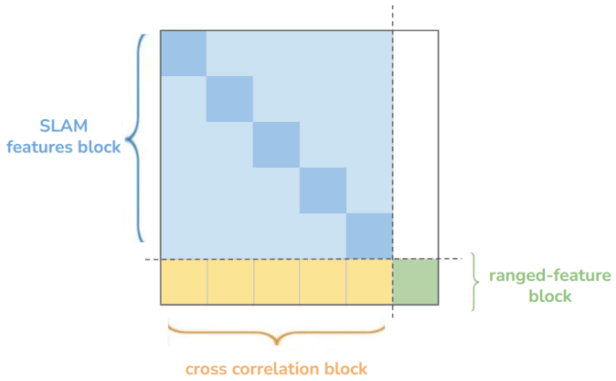



*Fig. 6 Block representation of the feature state covariance matrix*

A more robust real-world implementation of this RVIO code base would be to monitor the pixel distance between projected lidar rays and the chosen visual feature, and track the length of the feature before and after seeding depth information. While monitoring these metrics one could determine how strongly that feature reduces velocity and pose uncertainty in the filter update.

References

[1] https://github.com/david-dorf/spaceros_gz_demos/tree/main/spaceros_gz_demos

[2] H. Fjaer et al., “Flight Control System for NASA’s Mars Helicopter.” Online. Available: https://dartslab.jpl.nasa.gov/References/pdf/2019-mars-heli.pdf

[3] I. Alberico, J. Delaune, G. Cioffi, and D. Scaramuzza, “Structure-Invariant Range-Visual-Inertial Odometry,” arXiv.org, 2024. https://arxiv.org/abs/2409.04633 (accessed May 06, 2026).S. M. Metev and V. P. Veiko, Laser Assisted

[4] https://github.com/rpng/open_vins